ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C4 · FORCES AND MOTION

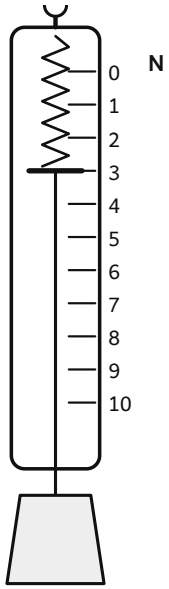
# Forces around us

★☆☆
Grade 4

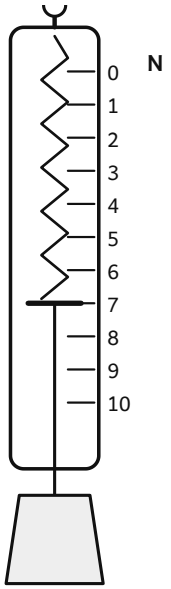
**Learning goals**

- Name forces.
- Use units.
- Show forces with arrows.

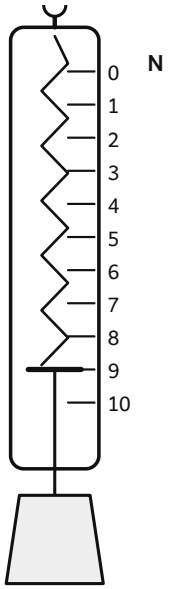
## 1 Read the forcemeters. [3]




*A*




*B*




*C*

| A | B | C |
|---|---|---|
| 3 N | 7 N | 9 N |

## 2 Name the force. [5]

| What happens | Force |
|---|---|
| a mango falls from a tree | gravity |
| a bicycle slows down when you brake | friction |
| a rope in tug-of-war is pulled tight | tension |
| a parachute falls slowly | air resistance |
| a boat floats | upthrust |

## 3 Sort the forces. [5]

SORT THESE

**friction** **a kick** **gravity** **tension** **magnetic force**

| Contact force | Non-contact force |
|---|---|
| friction | gravity |
| a kick | magnetic force |
| tension | |

## 4 Fill in the gaps. [4]

WORD BANK

**smaller** **newtons** **bigger** **metres** **kilograms** **same**

Mass is measured in kilograms. Weight is a force, so it is measured in newtons.

On the Moon your mass stays the same but your weight is smaller.

ANSWER KEY

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LEARN · PLAY · WONDER

THINK IT THROUGH · C4 · FORCES AND MOTION

# Forces in action

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Grade 4

**Learning goals**

- Draw force arrows.
- Explain motion using the laws of motion.

**1** **Draw arrows to show the forces on a boy on a slide: gravity pulling him down and friction slowing him.** [2]

Slide

Arrow pointing down (weight/gravity); arrow pointing up the slide, opposite to the movement (friction).

**2** **Explain each one using a law of motion.** [6]

| What happens | Explanation |
|---|---|
| a football rolling on grass slowly stops | law 1: friction (a force) acts on it and stops it |
| it is harder to push a full rickshaw van than an empty one | law 2: more mass needs a bigger force to change its speed |
| a swimmer pushes water back and moves forward | law 3: the water pushes the swimmer forward |

**3** **Why do people put sand or gravel on a slippery muddy path in the monsoon?** [2]

Sand increases friction so feet and tyres grip and do not slip.

**4** **In carrom, what happens to the striker's movement when it hits a coin?** [2]

The striker slows down and passes some of its movement (kinetic) energy to the coin, which starts to move.

ANSWER KEY

INVESTIGATE · C4 · FORCES AND MOTION

# Which surface has the most friction?

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**Learning goals**

- Plan a fair test.
- Measure distance.
- Explain results using friction.

Class 4D let a toy car roll down the same ramp onto four different surfaces and measured how far it travelled before stopping.

**1** **Complete the plan.** [3]

| | |
|---|---|
| Change | the surface |
| Measure | distance the car travels (cm) |
| Keep the same | the car, the ramp height and slope, where the car starts, no push |

**2** **Here are the results (mean of three tries). Draw a bar chart.** [3]

| Surface | tiles | wood | jute mat | sand paper |
|---|---|---|---|---|
| Distance (cm) | 180 | 140 | 70 | 35 |




**3** **Which surface had the most friction? How do you know?** [2]

Sandpaper: the car travelled the shortest distance, so friction slowed it most.

**4** **Why did they do each test three times and find the mean?** [1]

To make the results more reliable.

**5** **Which surface would be best for a playground slide landing area? Why?** [1]

A rough surface with lots of friction (like a jute mat) so children stop safely.

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REAL-WORLD CASE STUDY · C4 · FORCES AND MOTION

# Kabaddi and tug-of-war: the forces in our games

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**Learning goals**

- Find forces in real games.
- Use data from a tug-of-war.
- Think about fairness and safety.

Science | Physical education | Maths | Culture

**Kabaddi** is the national game of Bangladesh. A raider runs into the other team's half, tries to tag players and get back, while defenders try to hold them. Players **push**, **pull**, twist and change direction quickly. They play barefoot on soil or a mat, which gives good **friction** for gripping the ground.

In **tug-of-war** (*dori tana*), two teams pull a rope in opposite directions. The rope is in **tension**. If both teams pull with the same force, the forces are **balanced** and the rope does not move. If one team pulls harder, the forces are **unbalanced** and the rope moves towards the stronger team.

Example game data below.

**1** **Two teams pull a rope (example data).** [4]

| Round | Team Shapla (N) | Team Doel (N) | Which way does the rope move? |
|---|---|---|---|
| 1 | 800 | 800 | it does not move (balanced) |
| 2 | 900 | 750 | towards Shapla |
| 3 | 700 | 850 | towards Doel |

**a)** What is the difference in force in round 2?

900 − 750 = 150 N towards Shapla

**2** **Why do kabaddi players play barefoot on soil or a mat, not on smooth tiles?** [2]

Soil and mats have more friction, so their feet grip and they can push off, stop and turn without slipping.

**3** **When a raider pushes off the ground to run, which law of motion is at work? Explain.** [2]

Law 3: the feet push back on the ground and the ground pushes the raider forward.

**4** **Suggest two rules that make kabaddi or tug-of-war safe and fair.** [2]

e.g. teams of the same total mass; no pulling hair or clothes; a soft surface; gloves for the rope; a referee.

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REVIEW AND CHALLENGE · C4 · FORCES AND MOTION

# Show what you know

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**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

## 1 Find the key words.

| J | T | W | M | O | U | Y | H | F | H |
|---|---|---|---|---|---|---|---|---|---|
| I | V | E | A | F | M | V | J | O | C |
| C | L | I | S | R | O | J | G | G | L |
| Y | J | G | S | I | T | T | U | R | F |
| E | P | H | F | C | I | A | P | A | F |
| F | I | T | P | T | O | F | U | V | O |
| C | S | R | U | I | N | E | L | I | R |
| V | J | D | S | O | T | K | L | T | C |
| J | L | C | H | N | A | B | Y | Y | E |
| N | E | W | T | O | N | I | J | H | P |

**force friction gravity mass motion newton pull push weight**

## 2 Which is a non-contact force? [1]

- ○ friction
- ○ tension
- ● **gravity**
- ○ a push

## 3 What unit is used for weight? [1]

- ○ kilograms
- ● **newtons**
- ○ metres
- ○ litres

## 4 Draw a ball being kicked. Use arrows to show the kick, gravity and air resistance. [3]

Kicked ball

Kick arrow in the direction of movement; gravity arrow down; air resistance arrow opposite to the movement.

**5** **Challenge. An astronaut has a mass of 60 kg. What is her mass on the Moon? Is her weight bigger or smaller there? Explain.** [2]

Her mass is still 60 kg (the amount of matter does not change). Her weight is smaller (about 6 times less) because the Moon's gravity is weaker.